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## **Java Basics**

### **Introduction**

Java is a high-level, object-oriented programming language developed by Sun Microsystems. It follows the principle of "Write Once, Run Anywhere" because Java programs run on the Java Virtual Machine (JVM).

### **Features of Java**

- Platform Independent
- Object-Oriented
- Secure
- Robust
- Portable
- Multithreaded
- Distributed
- Dynamic

### **JVM**

The Java Virtual Machine (JVM) executes Java bytecode. It enables Java applications to run on different operating systems without changing the source code.

### **JDK**

The Java Development Kit (JDK) contains:

- Java Compiler (javac)
- JVM
- Java Runtime Environment (JRE)
- Development tools

### **JRE**

The Java Runtime Environment provides the libraries and JVM required to run Java applications.

### **OOP Concepts**

Java is based on four Object-Oriented Programming principles:

## **Encapsulation**

Wrapping data and methods into a single unit called a class.

## **Inheritance**

Allows one class to acquire properties and methods of another class.

## **Polymorphism**

Allows methods to perform different tasks based on the object calling them.

## **Abstraction**

Hides implementation details and exposes only essential functionality.

## **Exception Handling**

Java handles runtime errors using try, catch, finally, throw, and throws keywords.

## **Multithreading**

Multithreading allows multiple threads to execute simultaneously, improving performance and CPU utilization.

## **Advantages**

- Easy to learn
- Secure
- Cross-platform
- Large community support
- Automatic garbage collection